

What the metrics in `rabi-truncation` measure

A source-grounded route from finite-state fidelities to photon scattering

Technical note for `rabi-truncation` and `local_project`

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Abstract

The principal output of `rabi-truncation` is agreement between finite, time-evolved state vectors, not a transmission or reflection coefficient. This note defines every Hilbert-space label, derives the three plotted coupling-sweep quantities from their source lines, tests their mathematical interpretation with explicit counterexamples, and relates state agreement to directional one-photon probabilities. It also distinguishes finite-time bare-basis projections from asymptotic scattering amplitudes in a model without the rotating-wave approximation (RWA). A finite `full+totalcap` calculation is called a *reference calculation*, never an exact continuum solution. All source line numbers refer to the audited checkout; no files in the simulation code are changed by this note.

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1 Objects, indices, and normalization: the dictionary for the entire note

We use $\hbar = c = 1$ in mathematical derivations unless they are explicitly restored. The code keeps both constants in `Config`; its defaults equal one (`src/utilities.py:135--137`). A photon mode is labelled by an integer $m \in \{1, \dots, M\}$ and a *signed* wavevector $k_m \in \mathbb{R} \setminus \{0\}$. Its free energy is $\omega_m = |k_m|$, while the sign of k_m identifies propagation direction when the dispersion is $\omega = |k|$ and the mode convention is respected. The actual M is `config.modes` *after* the zero mode and any selected modes are removed (`src/utilities.py:162--184`). A mode label 'n3' means the occupation of the *third entry in this actual array*; it is not an immutable physical wavevector across configurations.

Symbol	Definition
N	<code>config.excitation_cap</code> ; its constraint depends on the chosen basis. It does not count the TLS excitation in any of the three field truncations.
$\mathbf{n} = (n_1, \dots, n_M)$	Field Fock occupations, $n_m \in \mathbb{N}_0$. The total <i>bare field</i> photon number is $\nu(\mathbf{n}) = \sum_{m=1}^M n_m$.
$s \in \{g, e\}$	Bare TLS ground/excited label; $\epsilon_g = 0$, $\epsilon_e = 1$. The bare atomic transition energy is ω_a (attribute <code>config.w_atom</code>).
$ \mathbf{n}; s\rangle$	Tensor product $ n_1, \dots, n_M\rangle \otimes s\rangle$. This is a bare basis state, not an interacting scattering eigenstate.
$\mathcal{B}_X$	Allowed field occupation set for $X \in \{\text{truncated}, \text{total}, \text{full}\}$. Its projector in the field-TLS space is $P_X = \sum_{\mathbf{n} \in \mathcal{B}_X, s} \mathbf{n}; s\rangle \langle \mathbf{n}; s $. The auxiliary basis has one additional index n_a .
d_X	Dimension $2 \mathcal{B}_X $; the stored ket has d_X complex entries.
$C_{\mathbf{n}s}^X(t)$	Expansion coefficient $\langle \mathbf{n}; s \psi_X(t) \rangle$ of one finite simulation. It is zero by extension when $\mathbf{n} \notin \mathcal{B}_X$.
$\rho^X(t)$	Global pure-state density operator $ \psi_X(t)\rangle \langle \psi_X(t) $ used only as mathematical notation; the program propagates the ket.
ρ_a^X, ρ_f^X	Partial traces $\text{Tr}_f \rho^X$ (a 2×2 TLS matrix) and $\text{Tr}_a \rho^X$ (a field matrix).
A, B	Two finite simulations being compared. In the coupling sweep, A is a candidate and B is <code>full+totalcap</code> , with the same nominal coupling g .
j, J, t_j	Output-time index, number of stored kets, and actual value in <code>sim.times[j]</code> . This j is unrelated to mode index m .
Π	An orthogonal projector representing a specified measurement event. Probabilities are $\langle \psi \Pi \psi \rangle$, not squared state-to-state fidelities unless Π is a rank-one projector onto the comparison state.

Every physically normalized ket satisfies

$$|\psi_X(t)\rangle = \sum_{\mathbf{n} \in \mathcal{B}_X} \sum_{s=g,e} C_{\mathbf{n}s}^X(t) |\mathbf{n}; s\rangle, \quad \sum_{\mathbf{n}, s} |C_{\mathbf{n}s}^X(t)|^2 = 1. \quad (1)$$

The `State` subclasses store adjacent g, e amplitudes; hence array entries $v[2i]$ and $v[2i+1]$ are the two amplitudes for the i th occupation. This is the index convention used by `atom_density_matrix` (`src/states.py:232--240`). `Simulation.time_evolve` calls `qutip.sesolve` on a ket and sets `normalize_output=True` (`src/simulation.py:27--33`). Thus the solver is a pure-state calculation; a 2×2 density matrix later used for the TLS is a *reduction* of that ket, not a global master-equation evolution. Solver output normalization masks raw norm drift; the `State.from_vector` wrapper normalizes each returned vector again before computing fidelities (`src/states.py:199--212`).

2 Which finite Hilbert spaces are compared?

Write $\mathbf{e}_m$ for the occupation vector with one in slot m and zero elsewhere. The four names in the audited repository correspond to

$$\mathcal{B}_{\text{truncated}} = \{\mathbf{0}\} \cup \{n\mathbf{e}_m : 1 \leq m \leq M, 1 \leq n \leq N\}, \quad d_{\text{truncated}} = 2(1 + MN), \quad (2)$$

$$\mathcal{B}_{\text{total}} = \{\mathbf{n} \in \mathbb{N}_0^M : \nu(\mathbf{n}) \leq N\}, \quad d_{\text{total}} = 2 \binom{M+N}{N}, \quad (3)$$

$$\mathcal{B}_{\text{full}} = \{\mathbf{n} \in \mathbb{N}_0^M : 0 \leq n_m \leq N \ \forall m\}, \quad d_{\text{full}} = 2(N+1)^M. \quad (4)$$

The fourth name, `truncated+atom`, appends an oscillator integer $n_a \in \{0, \dots, N\}$ *independently* to every $\mathbf{n} \in \mathcal{B}_{\text{truncated}}$; thus $d_{\text{aux}} = 2(1 + MN)(N+1)$. This oscillator is distinct from the two-state TLS index s . The formulas follow directly from `src/states.py:40--124`. In the comparisons of primary interest,

$$\mathcal{B}_{\text{truncated}} \subset \mathcal{B}_{\text{total}} \subset \mathcal{B}_{\text{full}} \quad (M \geq 2, N \geq 2). \quad (5)$$

The first inclusion is strict because $(1, 1, 0, \dots)$ exists in the total-cap basis but not in the one-occupied-mode basis. For $M = 2, N = 2$ the respective field sets have 5, 6, 9 elements. `full+totalcap` is the sweep's computational reference (`scripts/sweep_g_fidelity.py:43--46`); it is *not* an untruncated Hilbert space, and it does not include continuum momenta.

The auxiliary implementation has a particularly sharp consequence for its metric curves. In `HamiltonianAtom.interaction`, transitions that change n_a have `trans_index==M`, but the bosonic factor is evaluated from the field key `f'n{trans_index+1}'` (`src/hamiltonians.py:217--230`). This key is absent in those labels, so both occupations read as zero and the matrix element receives $\sqrt{\max(0, 0)} = 0$. The initial preparations occupy $n_a = 0$ (`src/states.py:312--334`). Therefore

$$[H_{\text{aux}}, P_{n_a=0}] = 0, \quad |\psi_{\text{aux}}(0)\rangle \in \text{ran } P_{n_a=0}, \quad qquad |\psi_{\text{aux}}(t)\rangle \in \text{ran } P_{n_a=0}. \quad (6)$$

Within this initial sector the evolution is the ordinary `truncated` evolution, up to basis ordering and numerical integration differences. Near-identical auxiliary and ordinary truncated fidelity curves therefore provide no independent evidence that an extra physical mode repairs the truncation.

3 The finite dynamics and the prepared packet

The three field bases use a projected Rabi-type Hamiltonian. Restoring the code's $\hbar, c$, its bare free part is

$$H_0 = \hbar c \sum_m |k_m| a_m^\dagger a_m + \hbar \omega_a |e\rangle\langle e|, \quad (7)$$

and the interaction has matrix elements of the form

$$V = \hbar \sum_m g \sqrt{|k_m|} (e^{-ik_m x_a} a_m + e^{ik_m x_a} a_m^\dagger) (\sigma^+ + \sigma^-), \quad H_X = P_X (H_0 + V) P_X. \quad (8)$$

Here x_a is `config.x_atom`, g is `config.g`, $\sigma^+ = |e\rangle\langle g|$, $\sigma^- = |g\rangle\langle e|$. The placement of the phase between a_m and $a_m^\dagger$ follows the matrix convention `H[ket, bra]` and `transition_sign` (`src/hamiltonians.py:39--88, 98--155, 232--255`); a rephasing of individual a_m changes this convention but not probabilities. The `RWA=True` branch retains only $a_m \sigma^+ + a_m^\dagger \sigma^-$ matrix elements (`src/hamiltonians.py:62--78`). The audited coupling $g \sqrt{|k_m|}$ does not contain an explicit $L^{-1/2}$ factor, so a comparison across lengths must specify how g scales with L if a fixed continuum spectral density is intended. Equation (7) matches the default $\hbar = 1$. If one changes `config.hbar`, the `truncated` and auxiliary free builders add the TLS term as `atom_index*w_atom` without $\hbar$, whereas the `full` and `total-cap` builders multiply it by $\hbar$ (`src/hamiltonians.py:98--110, 128--141, 160--174, 232--242`). Such a run would no longer compare one identical physical Hamiltonian across truncations.

The nominal number-state packet is

$$\tilde{c}_m = \exp\left[-\frac{\sigma_p^2}{2}(k_m - k_0)^2\right] e^{-ik_m x_p}, \quad c_m = \frac{\tilde{c}_m}{(\sum_\ell |\tilde{c}_\ell|^2)^{1/2}}, \quad \sigma_p > 0, \quad k_0, x_p \in \mathbb{R}. \quad (9)$$

Here σ_p, k_0, x_p are `sigma_photon`, `k_photon`, `x_photon`. This formula is `State.calculate_ck` (`src/states.py:181--185`). For `NumberState(1)` and initial TLS ground state, the preparation is

$$|\psi_X(0)\rangle = \sum_{m=1}^M c_m |\mathbf{e}_m; g\rangle \quad \text{for each basis containing all one-photon states.} \quad (10)$$

Crucially, Eq. (9) does *not* set $c_m = 0$ for $k_m < 0$. Its initial left-moving weight is

$$P_{\leftarrow}^{\text{in}} = \sum_{m:k_m < 0} |c_m|^2. \quad (11)$$

For the default $L = 10\pi$, $k_0 = 1$, $\sigma_p = 1$, and requested mode counts 16 and 32, evaluation of the *default* grid formula gives respectively $M = 15, 31$ and $P_{\leftarrow}^{\text{in}} \simeq 0.07404, 0.06188$. These numbers are conditional on those defaults; the committed result archives do not store a complete configuration. A raw left-moving population at late time is not automatically a reflection probability when Eq. (11) is nonzero.

For a product coherent input, the unprojected field state is

$$|\alpha\rangle = \bigotimes_{m=1}^M |\alpha c_m\rangle, \quad \langle \mathbf{n} | \alpha \rangle = e^{-|\alpha|^2/2} \prod_m \frac{(\alpha c_m)^{n_m}}{\sqrt{n_m!}}, \quad \sum_m |c_m|^2 = 1. \quad (12)$$

Each basis retains the coefficients in its own set and then normalizes the projected vector (`src/states.py:251--305, 307--358`). Thus coherent preparations in different truncations are generally *different physical initial kets*, even at $g = 0$.

4 Inventory: every metric family and what is actually plotted

Reported quantity	Mathematical object	Source	Can it alone yield $t(k), r(k)$?
state	$F_{\text{state}}(t_j) = \langle \psi_A(t_j) \psi_B(t_j) \rangle ^2$	<code>src/fidelities.py:26--53</code> ; sweep lines 71–99	No; it compares two full outputs for one preparation.
atom	Squared Uhlmann fidelity $F_a(t_j)$ between two 2×2 TLS reductions	<code>scripts/sweep_g_fidelity.py:60--68</code>	No; it discards all field detail.
photon_bound	$Q(t_j) = F_{\text{state}}(t_j)/F_a(t_j)$ when $F_a > 0$	<code>scripts/sweep_g_fidelity.py:71--79</code>	No; it is neither a field fidelity nor a bound.
P_e(t)	$\langle \psi(t) (I_f \otimes e\rangle\langle e) \psi(t) \rangle$	<code>src/simulation.py:44--63</code> ; <code>scripts/atom_evolution.py</code>	No; TLS population is not a directional field measurement.
entropy, purity	$-\text{Tr } \rho_a \log \rho_a$ and $\text{Tr } \rho_a^2$	<code>src/simulation.py:65--71</code> ; <code>src/utilities.py:247--255</code>	No; they characterize TLS-field entanglement for a pure global state.
energyprofile	Partition of $\langle H \rangle$ by mode or bare photon count	<code>src/energy_profile.py:47--146</code>	No; energy contribution is not channel probability or flux.
cumulativefidelity	$F_{N,N+1}(T)$ between adjacent total caps	<code>scripts/full_cumulative_fidelity.py:19--69</code>	No; it checks one sequence of finite models.
mode selection	F after embedding selected k modes in the full grid	<code>src/fidelities.py:56--151</code> ; mode-selection script	No; it checks an approximation to a reference trajectory.

The coupling-sweep plot stores the *sampled-time mean* of the first three quantities, not their final-time values, and no principal fidelity script computes T or R . The exact definitions, conditions, and counterexamples follow.

5 Global state fidelity: exact meaning and the basis-alignment hazard

For normalized $|\psi_A\rangle, |\psi_B\rangle$ in one common physical Hilbert space, define

$$F_{\text{state}}(A, B) = |\langle \psi_A | \psi_B \rangle|^2 = \left| \sum_{\mathbf{n} \in \mathcal{B}_A \cap \mathcal{B}_B} \sum_{s=g,e} (C_{\mathbf{n}s}^A)^* C_{\mathbf{n}s}^B \right|^2. \quad (13)$$

The second equality extends both states by zero outside their finite supports. The *complex* overlap is summed before taking its modulus squared: replacing it by $\sum |C^A|^2 |C^B|^2$ would erase interference phases. The value is between zero and one by Cauchy–Schwarz. It equals one if and only if the two normalized kets differ by one global phase. At a fixed time, it is also the probability that a projective test for *the entire reference ket* accepts the candidate ket. It is not the probability of an outgoing photon on either side.

The implementation selects a basis class with `common_basis`, loops over its labels, accumulates the complex inner product, and squares its modulus (`src/fidelities.py:8--53`):

```
1 common = common_basis(state1, state2)
2 for basis_element in ref.all_states():
3     fidelity += np.conj(state1[basis_element]) * state2[basis_element]
4 fidelity = np.abs(fidelity) ** 2
```

For `truncated` against `full+totalcap` on an identical grid, the enumerated set is $\mathcal{B}_{\text{truncated}} = \mathcal{B}_A \cap \mathcal{B}_B$, so Eq. (13) is correct. For `truncated+atom` against `full+totalcap`, the code’s `common` `StateTruncated` enumeration corresponds to $n_a = 0$; this is sufficient for the prepared invariant sector of Eq. (6).

The word “universal” in the `common_basis` docstring is too strong. Whenever the types differ, it chooses `StateTruncated` (`src/fidelities.py:16--24`). In particular,

$$\mathcal{B}_{\text{full}} \cap \mathcal{B}_{\text{total}} = \mathcal{B}_{\text{total}} \supsetneq \mathcal{B}_{\text{truncated}}. \quad (14)$$

Set $M = 2, N = 2$ and let both states be the *same* ket $|1, 1; g\rangle$, one represented as `StateFull`, the other as `StateTotalCap`. The physical answer is $F_{\text{state}} = 1$. The actual function iterates only vacuum and one-occupied-mode labels, finds zero overlap there, and returns 0. A direct small-grid probe of the audited checkout produced exactly this $1 \mapsto 0$ discrepancy. Hence the cross-type routine is valid only when its enumerated labels cover the true support intersection, or when all omitted common amplitudes vanish. The primary truncated-versus-total-cap sweeps meet that condition; an arbitrary heterogeneous comparison does not.

There is a second, independent requirement: mode index m must name the same k_m in both states. Raw `fidelity_statevector` does not check this. For selected grids, `physical_mode_map` aligns modes by k using `np.isclose`, and `embed_in_grid` inserts vacuum in missing modes (`src/fidelities.py:56--115`); `fidelity_to_reference` uses that embedding before Eq. (13) (lines 138–152). A comparison at unlike physical k arrays without this map has no well-defined interpretation as an overlap of the same field modes. The embedding itself is isometric only if all selected modes are distinct matches and the basis label mapping is injective, as intended by the subset construction.

5.1 What a high or low state fidelity can say about scattering

At one *outgoing* time, high F_{state} is strong evidence that two simulations agree on *all* bounded measurement probabilities, including any identically defined transmission projector; Section 15 gives the sharp quantitative inequality. Low F_{state} does not identify which measurement differs. For instance, with two orthogonal right-moving one-photon modes,

$$|A\rangle = \frac{|k_1; g\rangle + |k_2; g\rangle}{\sqrt{2}}, \quad |B\rangle = \frac{|k_1; g\rangle - |k_2; g\rangle}{\sqrt{2}}, \quad F_{\text{state}} = 0, \quad T_A = T_B = 1, \quad R_A = R_B = 0. \quad (15)$$

The relative spectral phase differs although integrated directional probabilities agree. Conversely, F_{state} compares two numerical answers, not either answer with an unknown exact physical S matrix. Even $F_{\text{state}} = 1$ only establishes that those two finite calculations coincide for the specified input and time.

6 Reduced TLS matrix, excitation, purity, and entropy

Write $|\psi\rangle = |\phi_g\rangle|g\rangle + |\phi_e\rangle|e\rangle$, where $|\phi_s\rangle = \sum_{\mathbf{n}} C_{\mathbf{n}s}|\mathbf{n}\rangle$ may be unnormalized. Tracing over the field gives the explicitly defined 2×2 matrix

$$\rho_a = \text{Tr}_f |\psi\rangle\langle\psi| = \begin{pmatrix} \sum_{\mathbf{n}} |C_{\mathbf{n}g}|^2 & \sum_{\mathbf{n}} C_{\mathbf{n}g} C_{\mathbf{n}e}^* \\ \sum_{\mathbf{n}} C_{\mathbf{n}e} C_{\mathbf{n}g}^* & \sum_{\mathbf{n}} |C_{\mathbf{n}e}|^2 \end{pmatrix}, \quad \text{Tr } \rho_a = 1. \quad (16)$$

This is precisely the even/odd slicing and dot products in `src/states.py:228--240`. The off-diagonal entry is a coherent overlap between two *field* vectors. Its absence may result from entanglement, not from a thermal or dissipative environment.

The bare TLS excitation probability used by `compute_excited_probability` is

$$P_e(t) = (\rho_a(t))_{ee} = \sum_{\mathbf{n}} |C_{\mathbf{n}e}(t)|^2 = \langle\psi(t)|(I_f \otimes |e\rangle\langle e|)|\psi(t)\rangle. \quad (17)$$

The function returns the whole series when `t=None`, the last stored value when `t=-1`, or an index chosen as `int(t/config.dt)` otherwise (`src/simulation.py:44--63`). Because the actual `sim.times` spacing is generally not `config.dt`, the numeric `t` branch is an *indexing approximation*; it should not be read as an exact interpolation at the requested physical time. The `atom_evolution.py` figure plots $\Re P_e(t_j)$ (`scripts/atom_evolution.py:93--105`).

For eigenvalues $\lambda_1, \lambda_2 \geq 0$ of ρ_a with $\lambda_1 + \lambda_2 = 1$, the code's optional diagnostics are

$$\gamma_a = \text{Tr } \rho_a^2 = \lambda_1^2 + \lambda_2^2 = 1 - 2 \det \rho_a, \quad S_a = -\text{Tr}(\rho_a \log \rho_a) = -\sum_{r=1}^2 \lambda_r \log \lambda_r. \quad (18)$$

Thus $1/2 \leq \gamma_a \leq 1$, $0 \leq S_a \leq \log 2$ for normalized pure global kets. Example:

$$|\psi\rangle = \frac{|\mathbf{0}; g\rangle + |\mathbf{e}_1; e\rangle}{\sqrt{2}} \implies \rho_a = \frac{I_2}{2}, \quad \gamma_a = \frac{1}{2}, \quad S_a = \log 2, \quad (19)$$

although $\rho = |\psi\rangle\langle\psi|$ is pure and evolves unitarily. The entropy is an entanglement measure of the TLS versus *all* retained field degrees of freedom for a pure global ket; it does not encode which side an outgoing photon occupies.

7 TLS fidelity: a precisely defined but deliberately coarse comparison

For two TLS matrices ρ_a^A, ρ_a^B , define the *root* Uhlmann fidelity

$$f_a = \text{Tr} \sqrt{\sqrt{\rho_a^A} \rho_a^B \sqrt{\rho_a^A}}, \quad F_a = f_a^2. \quad (20)$$

The sweep calls `qutip.fidelity(rho1, rho2)` and squares its result, so its `atom` array contains F_a , not f_a (`scripts/sweep_g_fidelity.py:60--68`). This convention makes $F_a = |\langle a | b \rangle|^2$ when the TLS reductions are pure kets $|a\rangle, |b\rangle$.

For *any* two 2×2 density matrices, Eq. (20) has an elementary closed form:

$$F_a = \text{Tr}(\rho_a^A \rho_a^B) + 2\sqrt{\det \rho_a^A \det \rho_a^B}. \quad (21)$$

To verify it, set $X = \sqrt{\rho_a^A} \rho_a^B \sqrt{\rho_a^A}$ with eigenvalues $\mu_{\pm}$. Then $(\text{Tr} \sqrt{X})^2 = \mu_+ + \mu_- + 2\sqrt{\mu_+ \mu_-}$, while $\text{Tr} X = \text{Tr}(\rho_a^A \rho_a^B)$ and $\det X = \det \rho_a^A \det \rho_a^B$. This derivation also fixes the square convention without relying on a library name.

A basic information-loss example is

$$\begin{aligned} |A\rangle &= |1_{k>0}; g\rangle, & |B\rangle &= |1_{k<0}; g\rangle, \\ \rho_a^A &= \rho_a^B = |g\rangle\langle g|, & F_a &= 1, \quad F_{\text{state}} = 0, \\ (T_A, R_A) &= (1, 0), & (T_B, R_B) &= (0, 1). \end{aligned} \quad (22)$$

All TLS-only observables, including P_e , purity, and entropy, agree perfectly, yet the scattering direction is opposite. Thus a high `atom` curve is compatible with a wrong reflection coefficient. It does bound differences in *TLS* measurement probabilities, as shown in Section 15.

8 The plotted photon_bound: quotient, not photon fidelity

The sweep constructs three time series (`scripts/sweep_g_fidelity.py:71--79`):

$$\begin{aligned} F_s(j) &= F_{\text{state}}(\psi_A(t_j), \psi_B(t_j)), \\ F_a(j) &= F_a(\rho_a^A(t_j), \rho_a^B(t_j)), \\ Q(j) &= \begin{cases} F_s(j)/F_a(j), & F_a(j) > 0, \\ \text{NaN}, & F_a(j) = 0. \end{cases} \end{aligned} \quad (23)$$

The dictionary key and plot filename `photon_bound` refer to Q . Neither the formula nor the code computes the field reduced-state fidelity F_f defined in Section 9. `np.where` handles a zero denominator by storing `NaN`; a later `np.nanmean` omits those times. In exact arithmetic, monotonicity of fidelity under the partial trace implies $F_s \leq F_a$, so $0 \leq Q \leq 1$ when defined. This inequality makes Q look like a fidelity, but does not make it one.

There is one useful special case. If *both* states factorize as $|\phi_A\rangle|a_A\rangle$ and $|\phi_B\rangle|a_B\rangle$, then

$$F_s = |\langle \phi_A | \phi_B \rangle|^2 |\langle a_A | a_B \rangle|^2 = F_f F_a, \quad Q = F_f \quad \text{when } F_a > 0. \quad (24)$$

Once the TLS and field are entangled in either calculation, the factorization of the *overlap* does not follow from the factorization of a Hilbert space. Dividing by F_a is not a partial trace and is not a conditional probability of transmission.

Two normalized counterexamples show that Q is neither a lower nor an upper bound on F_f . All kets below live in one two-dimensional field space $\text{span}\{|0\rangle, |1\rangle\}$ and one TLS space $\text{span}\{|g\rangle, |e\rangle\}$.

- (a) Take $|A\rangle = (|0; g\rangle + |1; e\rangle)/\sqrt{2}$ and $|B\rangle = (|0; g\rangle - |1; e\rangle)/\sqrt{2}$. Orthogonality gives $F_s = 0$. Both reductions equal $I_2/2$, so $F_a = F_f = 1$. Thus $Q = 0 < F_f$.
- (b) Take $|A\rangle = |0; g\rangle$ and $|B\rangle = \frac{1}{2}|0; g\rangle + \frac{1}{2}|0; e\rangle + 2^{-1/2}|1; e\rangle$. The three terms of B are orthogonal and their squared norms sum to one. Here $F_s = 1/4$. The g probability in B is $1/4$, so the pure-versus-mixed formula gives $F_a = 1/4$. The field-0 probability in B is $1/2$, so $F_f = 1/2$. Therefore $Q = 1 > F_f$.

No universal ordering between Q and F_f exists. In particular, the word “bound” in a caption or filename is not supported by the mathematics of Eq. (23). A further subtlety is that the plotted quantity is $\bar{Q}$, not $\bar{F}_s/\bar{F}_a$; these are generally different even when all denominators are positive.

9 The exact field fidelity that could be computed from the existing kets

The field reduced state is

$$\rho_f^X = \text{Tr}_a |\psi_X\rangle\langle\psi_X| = |\phi_g^X\rangle\langle\phi_g^X| + |\phi_e^X\rangle\langle\phi_e^X|, \quad \text{rank } \rho_f^X \leq 2. \quad (25)$$

`State.photon_density_matrix` explicitly allocates this field matrix from the two even/odd coefficient vectors (`src/states.py:217--226`), but the coupling sweep does not call it. Its dimension can be large even though its rank is at most two.

After embedding both finite states into the *same* field mode grid, define a 2×2 cross Gram matrix with TLS indices $s, s' \in \{g, e\}$:

$$G_{ss'} = \langle \phi_s^A | \phi_{s'}^B \rangle = \sum_{\mathbf{n} \in \mathcal{B}_A \cap \mathcal{B}_B} (C_{\mathbf{n}s}^A)^* C_{\mathbf{n}s'}^B. \quad (26)$$

Then the *exact squared Uhlmann field fidelity* is

$$F_f = \left(\text{Tr} \sqrt{\sqrt{\rho_f^A} \rho_f^B \sqrt{\rho_f^A}} \right)^2 = \|G\|_1^2 = (\sigma_1(G) + \sigma_2(G))^2, \quad (27)$$

where σ_1, σ_2 are the nonnegative singular values and $\|G\|_1$ is their sum. For a derivation, form coefficient maps $C_X : \mathbb{C}^2 \rightarrow \mathcal{H}_f$ with columns $|\phi_g^X\rangle, |\phi_e^X\rangle$. Then $\rho_f^X = C_X C_X^\dagger$ and $G = C_A^\dagger C_B$. Polar decomposition of C_A, C_B , or the nonzero singular-value identity for $C_A C_A^\dagger$ and $C_A^\dagger C_A$, reduces the trace norm in the first expression to $\|C_A^\dagger C_B\|_1$. Thus a 2×2 singular-value decomposition suffices: a dense field density matrix is not mathematically required. This is the computation implemented in the independent `codex/code/src/fidelities.py:7--44`, although that prototype is separate from the audited repository.

The exact inequalities for properly aligned and normalized global states are

$$0 \leq F_s \leq F_a \leq 1, \quad 0 \leq F_s \leq F_f \leq 1. \quad (28)$$

They express the fact that discarding degrees of freedom cannot make two states *more distinguishable* by fidelity. There is no universal ordering between F_a and F_f . Equation (27) compares *entire field states*, including vacuum, left/right components, spectral shape, multiphoton components and their coherence. It is therefore closer to a field question than F_a , but it is still a comparison of two calculations, not the physical transmission or reflection of either one.

10 The temporal statistic on the coupling-sweep plots

The solver stores

$$J = \text{int}(T/\delta), \quad t_j = \frac{jT}{J-1}, \quad j = 0, \dots, J-1, \quad T > 0, \delta > 0, \quad (29)$$

for $J \geq 2$, where T is `config.t` and δ is `config.dt`, because the source is `np.linspace(0, config.t, int(config.t/config.dt))` (`src/simulation.py:27--33`). The actual spacing is $\Delta t_{\text{out}} = T/(J-1)$, generally *not* δ . For example $T = 10, \delta = 0.1$ gives $J = 100$ and $\Delta t_{\text{out}} = 10/99$. Here δ determines the requested output *count*; the adaptive BDF solver still chooses internal integration steps independently. The fidelity calculations pair states by their list order (`zip`), not by checking equality of physical time arrays (`scripts/sweep_g_fidelity.py:71--79`). In the sweep the configurations share T, δ , so that pairing is aligned.

For each coupling g_i and candidate scheme X , the stored point is

$$\overline{F}_s(g_i; X) = \frac{1}{J} \sum_{j=0}^{J-1} F_s^{X, \text{ref}}(t_j; g_i), \quad \overline{F}_a(g_i; X) = \frac{1}{J} \sum_j F_a^{X, \text{ref}}(t_j; g_i), \quad (30)$$

$$\overline{Q}(g_i; X) = \frac{1}{|I_i|} \sum_{j \in I_i} \frac{F_s^{X, \text{ref}}(t_j; g_i)}{F_a^{X, \text{ref}}(t_j; g_i)}, \quad I_i = \{j : F_a^{X, \text{ref}}(t_j; g_i) > 0\}. \quad (31)$$

These equations are `np.mean` and `np.nanmean` in `scripts/sweep_g_fidelity.py:81--99`. Equal weights are assigned to all saved times, including endpoints. They approximate a time integral only when J is sufficiently large for the variation being studied, and they deliberately include the incoming and interacting stages. They are not a fidelity of a time-averaged state and not an outgoing-time observable. A large $\overline{F}_s$ does not ensure a large $F_s(T)$: $F_s(t_j) = 1$ for $J-1$ samples and $F_s(t_{J-1}) = 0$ still gives $\overline{F}_s = 1 - 1/J$.

10.1 Why coherent curves can start below one at zero coupling

Let $A = |\alpha|^2$ and $p_m = |c_m|^2$, so $\sum_m p_m = 1$. The unprojected coherent state's squared norm in the total-cap basis is the Poisson cumulative probability

$$Z_{\text{total}} = \|P_{\text{total}}|\alpha\rangle\|^2 = e^{-A} \sum_{r=0}^N \frac{A^r}{r!}. \quad (32)$$

The one-occupied-mode basis retains vacuum and occupations $n\mathbf{e}_m$ only, so

$$Z_{\text{truncated}} = e^{-A} \left[1 + \sum_{m=1}^M \sum_{n=1}^N \frac{(Ap_m)^n}{n!} \right]. \quad (33)$$

Since $P_{\text{truncated}}P_{\text{total}} = P_{\text{truncated}}$, the overlap of the *separately normalized* preparations is

$$F_s(0) = \frac{Z_{\text{truncated}}}{Z_{\text{total}}}. \quad (34)$$

For $g = 0$, both projected free Hamiltonians act identically on their common occupation states and preserve these subspaces, so this initial discrepancy persists at every time: the nonunit value is a preparation/projection effect, not an interaction-induced scattering error. For an initial `NumberState(1)` the state lies in both bases, and $F_s(0) = 1$. For a coherent state, Eq. (34) can be substantially below one even at $g \rightarrow 0$. That statement is directly relevant to the committed coherent sweeps.

10.2 What the committed numeric archives establish

The four readable `results/results*/fidelity_vs_g.npz` files each contain 50 couplings from 10^{-5} to 0.2 and six arrays: $\overline{F}_s, \overline{F}_a, \overline{Q}$ for each of `truncated` and `truncated+atom`. The following are direct values read from the `truncated` arrays; the adjacent auxiliary arrays agree to the shown precision. “First” and “last” denote archive order along g , not time:

Archive	$\overline{F}_s$ first $\rightarrow$ last	$\overline{F}_a$ first $\rightarrow$ last	$\overline{Q}$ first $\rightarrow$ last
<code>results_16</code>	1.000000 $\rightarrow$ 0.647275	1.000000 $\rightarrow$ 0.937931	1.000000 $\rightarrow$ 0.688626
<code>results_32</code>	1.000000 $\rightarrow$ 0.557383	1.000000 $\rightarrow$ 0.870746	1.000000 $\rightarrow$ 0.638128
<code>results_coherent_16</code>	0.763935 $\rightarrow$ 0.396654	1.000000 $\rightarrow$ 0.984470	0.763935 $\rightarrow$ 0.402276
<code>results_coherent_32</code>	0.765522 $\rightarrow$ 0.158069	1.000000 $\rightarrow$ 0.974237	0.765522 $\rightarrow$ 0.161630

These are finite-model state-agreement data. The archives contain the g axis but no full configuration, signed k list, time traces or outgoing channel projections. The filenames suggest number versus coherent preparations and requested mode counts, but those suggestions do not prove the exact command lines or solver settings used to generate them. In the coherent archives, the small- g state-fidelity deficit is consistent with Eq. (34); the TLS fidelity near one is consistent with Eq. (22)’s warning that the TLS can remain almost unchanged while field states differ. None of these six columns is a measured $T(k)$, $R(k)$ or complex scattering amplitude.

11 Two further comparisons: cap convergence and mode selection

11.1 Adjacent total-photon caps

The script `scripts/full_cumulative_fidelity.py:19--69` independently solves the `full+totalcap` model at $N = 1, \dots, N_{\text{max}}$, takes each final stored ket, and plots

$$F_{N,N+1}(T) = |\langle \psi_N(T) | \psi_{N+1}(T) \rangle|^2, \quad |\psi_N(T)\rangle \in \text{ran } P_N \subset \text{ran } P_{N+1}. \quad (35)$$

The expression uses the smaller basis labels for the overlap; that is correct because $P_N P_{N+1} = P_N$. But the script’s *coherent* initial preparations are separately projected and normalized. Consequently

$F_{N,N+1}(0)$ can already be less than one, and final discrepancy contains both initial-projection and dynamical-cutoff contributions. Even a sequence with $F_{N,N+1}(T) \rightarrow 1$ at several accessible N is evidence only of adjacent-cap stability at that fixed M, L, T, g and that chosen input. It is neither a bound on error versus the infinite-cap limit without a summable tail estimate nor evidence of convergence in the momentum discretization. The script catches exceptions and stops the loop, so missing larger- N points are computational absences, not fidelity values.

A useful exact decomposition in the common $(N + 1)$ -cap space is

$$\begin{aligned} |\psi_N(T)\rangle - |\psi_{N+1}(T)\rangle &= U_N(T) [|\psi_N(0)\rangle - P_N |\psi_{N+1}(0)\rangle] \\ &\quad + [U_N(T)P_N - P_N U_{N+1}(T)] |\psi_{N+1}(0)\rangle \\ &\quad - (I - P_N) U_{N+1}(T) |\psi_{N+1}(0)\rangle. \end{aligned} \quad (36)$$

Here $U_N = e^{-iH_N T}$ acts on $\text{ran } P_N$; U_{N+1} acts on the larger space. The three lines are, respectively, an initial mismatch *within* the smaller basis, a difference of projected dynamics within that basis, and amplitude outside it. At $T = 0$ the middle line vanishes, while the last line is the initial omitted tail. One cannot attribute final infidelity solely to altered dynamics without measuring the preparation and outside-sector contributions.

11.2 Selected momentum modes

For selected set $K_{\text{sel}} \subset K_{\text{ref}}$, `physical_mode_map` computes an index injection ι satisfying $k_j^{\text{sel}} \simeq k_{\iota(j)}^{\text{ref}}$ (`src/fidelities.py:56--79`). The field embedding J_ι maps

$$J_\iota |n_1, \dots, n_{M_{\text{sel}}}\rangle = |N_1, \dots, N_{M_{\text{ref}}}\rangle, \quad N_{\iota(j)} = n_j, \quad N_{m \notin \text{im } \iota} = 0. \quad (37)$$

It preserves norm when the mapping is one-to-one. `embed_in_grid` performs this relabelling, and `fidelity_to_reference` then calls the global overlap (`src/fidelities.py:82--151`). The `compare_mode_selection.py` sweep and heatmaps plot a time mean of this fidelity against the single unselected `full+totalcap` reference (`scripts/compare_mode_selection.py:75--121, 150--169`). They vary both occupation restrictions and retained momenta, and the heatmaps vary two window widths. Their colors represent *state agreement*, not transmitted intensity; their “ground truth” is the finite reference named in code.

For a one-photon packet of Eq. (10), simply projecting to selected modes and normalizing changes the initial state. Define $p_{\text{keep}} = \sum_{m: k_m \in K_{\text{sel}}} |c_m|^2$. If the selected preparation uses the same unnormalized envelope restricted to K_{sel} , then

$$|\psi_{\text{sel}}(0)\rangle = \frac{1}{\sqrt{p_{\text{keep}}}} \sum_{k_m \in K_{\text{sel}}} c_m |1_m; g\rangle, \quad F_s(0) = p_{\text{keep}}. \quad (38)$$

A fidelity deficit at time zero therefore quantifies a known discarded *input* fraction before any TLS scattering. For coherent states, omitted modes initially carry independent coherent amplitudes; the untruncated overlap with their vacuum is $\exp[-\sum_{m \notin K_{\text{sel}}} |\alpha c_m|^2]$ after squaring, with additional changes from finite Fock projection. Reporting $F_s(0)$ or p_{keep} is necessary to separate this preparation effect from mode-selection dynamics.

12 Energy diagnostics are not directional photon probabilities

The repository adds two exact decompositions of the *expectation* $E(t) = \langle \psi(t) | H | \psi(t) \rangle$ (`src/energy_profile.py:47--146`). Let q enumerate all TLS-resolved basis cells, let ψ_q be the ket component, let ν_q be the field photon number of cell q , and let $P_\nu = \sum_{q: \nu_q = \nu} |q\rangle \langle q|$. The excitation-bin quantity is

$$E_\nu(t) = \sum_{q: \nu_q = \nu} \Re[\psi_q^*(H\psi)_q] = \Re\langle \psi | P_\nu H | \psi \rangle = \frac{1}{2} \langle \psi | \{P_\nu, H\} | \psi \rangle, \quad \sum_\nu E_\nu = \langle \psi | H | \psi \rangle. \quad (39)$$

This is `np.real(np.conj(psi)*(H@psi))` followed by `np.bincount` (`src/energy_profile.py:130--137`). The bin label ν omits the TLS excitation but the energy assigned to that bin includes the TLS and interaction terms of those basis cells. Since $[P_\nu, H] \neq 0$ without RWA, E_ν is an energy partition, not the energy of an independently conserved photon-number sector. It can be negative. By contrast, the photon-number *probability*

$$p_\nu(t) = \langle \psi | P_\nu | \psi \rangle = \sum_{q: \nu_q = \nu} |\psi_q(t)|^2 \geq 0, \quad \sum_\nu p_\nu = 1, \quad (40)$$

is a distinct object and is not the displayed E_ν .

The mode partition is

$$E_m(t) = \hbar c |k_m| \langle n_m \rangle_t + \langle H_{\text{int},m} \rangle_t, \quad E_{\text{atom}}(t) = \langle H_{\text{atom,diag}} + H_{\text{int,atom}} \rangle_t, \quad E_{\text{atom}} + \sum_m E_m = \langle H \rangle_t. \quad (41)$$

Here $H_{\text{int},m}$ receives off-diagonal matrix elements that change field mode m (`src/energy_profile.py:86--114, 119--128`). The atom bin collects the remaining diagonal energy and any off-diagonal term not attributed to a field mode. This partition is internally checkable by its sum but is not unique as a *local energy density*; an interaction term's attribution is a convention. Its k_m axis is signed, but E_m includes interaction energy and is not a photon count, an energy current through a detector, or $|t(k_m)|^2$. The GIF and heatmap only animate or display these values (`scripts/energy_profile.py:105--219`). Exact constancy of $\langle H \rangle$ checks finite closed dynamics; it does not establish that a wavepacket has scattered or escaped the TLS.

13 The target observable: a scattering matrix, not a fidelity

13.1 Asymptotic channel definition

A scattering matrix maps *incoming channel amplitudes* to *outgoing channel amplitudes*. Its definition requires a channel Hilbert space $\mathcal{H}_{\text{ch}}$, a channel Hamiltonian H_{ch} , and an injection $J : \mathcal{H}_{\text{ch}} \rightarrow \mathcal{H}$ representing well-separated free photons with the scatterer in an appropriate stationary state. If the wave operators exist, set

$$W_{\text{in}} = \text{s} \lim_{t \rightarrow -\infty} e^{iHt} J e^{-iH_{\text{ch}}t}, \quad W_{\text{out}} = \text{s} \lim_{t \rightarrow +\infty} e^{iHt} J e^{-iH_{\text{ch}}t}, \quad S = W_{\text{out}}^\dagger W_{\text{in}}. \quad (42)$$

The *strong limit* means convergence on each channel vector in norm. Equation (42) fixes the in/out convention; S is an operator, not a single probability. In a finite periodic box with a discrete spectrum, persistent recurrences generally preclude a literal $t \rightarrow \infty$ scattering limit. A finite simulation can approximate a scattering experiment in a time window after the outgoing packet separates from the TLS and before it wraps around the box, followed by checks as L , mode bandwidth, and Fock cap change.

Without RWA, the bare product $|\mathbf{0}; g\rangle$ is generally *not* an eigenstate of H because $a_m^\dagger \sigma^+ |\mathbf{0}; g\rangle \neq 0$. The stationary scatterer for an exact asymptotic channel is a *dressed* ground state $|G\rangle$ of the interacting Hamiltonian, with its localized photon cloud, rather than the bare TLS ground label by itself. Formally, an elastic one-photon input channel is a distant incoming photon on $|G\rangle$, and an elastic output is a distant outgoing photon with the scatterer back in $|G\rangle$. At strong coupling, possible bound-state or inelastic channels must be included explicitly; see the primary research sources at the end of the note. The finite bare-basis coefficient $C_{e,m,g}$ is therefore an easily computed *proxy* for an elastic channel amplitude, not a complete non-RWA asymptotic definition.

There is a precise symmetry that helps categorize channels. Define $\sigma_z = |e\rangle\langle e| - |g\rangle\langle g|$ and $\Pi = \sigma_z(-1)^{\hat{N}_f}$, $\hat{N}_f = \sum_m a_m^\dagger a_m$. The Rabi interaction flips the TLS and changes $\hat{N}_f$ by one, so $[H, \Pi] = 0$ even though $[H, \hat{N}_f + |e\rangle\langle e|] \neq 0$. A bare $|1_m; g\rangle$ state and a bare three-photon $|3 \text{ photons}; g\rangle$ state have the same parity. Thus a one-to-three-photon channel is not excluded by parity; whether it is *open* depends on total energy, dispersion, band edges and the dressed spectrum. A bare one-photon probability need not exhaust asymptotic probability in a no-RWA calculation.

13.2 Complex transmission and reflection amplitudes

For a localized, stationary scatterer with asymptotic translation-invariant waveguide channels and linear even dispersion $\omega(k) = |k|$, let $|k; G\rangle \in \mathcal{H}_{\text{ch}}$ label a signed-momentum one-photon channel with the scatterer in its ground state. For incident $k > 0$ and signed outgoing $k' \neq 0$, one elastic S -matrix convention is

$$\langle k'; G | S | k; G \rangle = t(k) \delta(k' - k) + r(k) \delta(k' + k), \quad k > 0, \quad k' \in \mathbb{R} \setminus \{0\}. \quad (43)$$

Here $t(k), r(k) \in \mathbb{C}$ are *amplitudes*. The delta factors state energy conservation and the two elastic directions; if a different continuum normalization or dispersion is used, density-of-states factors must be included consistently. In a purely elastic two-channel situation, unitarity gives $|t(k)|^2 + |r(k)|^2 = 1$. If other asymptotic channels are open, it gives instead

$$|t(k)|^2 + |r(k)|^2 + \sum_{\chi \in \text{other channels}} P_{\chi}(k) = 1 \quad (44)$$

for unit incoming flux, with an integral in the sum when final energies are continuous. Equation (44) is not the same as a finite-time identity involving the bare $P_e(t)$.

For normalized incoming spectral envelope $f(k)$ supported on $k > 0$, the elastic outgoing amplitudes are, in the diagonal elastic approximation,

$$f_{\rightarrow}^{\text{out}}(k) = t(k)f(k), \quad f_{\leftarrow}^{\text{out}}(-k) = r(k)f(k), \quad k > 0. \quad (45)$$

Consequently the integrated elastic probabilities are

$$T_{\text{el}}[f] = \int_0^{\infty} |f(k)|^2 |t(k)|^2 dk, \quad R_{\text{el}}[f] = \int_0^{\infty} |f(k)|^2 |r(k)|^2 dk. \quad (46)$$

A wavepacket experiment measures these *spectral averages*, not $t(k_0)$ and $r(k_0)$ automatically. The narrowband approximation $T \simeq |t(k_0)|^2$, $R \simeq |r(k_0)|^2$ requires both f concentrated near k_0 and t, r slowly varying across its width. Even exact knowledge of T_{el} and R_{el} loses the phases of t and r ; those phases control interference and time delay. Reconstructing an S -matrix operator requires enough distinct incoming states or a direct channel-resolved resolvent calculation, not one integrated curve.

14 What local_project calls transmitted and reflected

The minimal local calculation explicitly zeros the negative- k entries of its incoming Gaussian before normalization (`local_project/simu_Fock_space/src/experiment.py:125--135`). Its one-photon coefficient arrays are $C_{\mathbf{e}_m, g}(t)$ and $C_{\mathbf{e}_m, c}(t)$ in its chosen picture and basis. In a common bare field-TLS space with no $k = 0$ mode, define

$$\Pi_{\rightarrow, 1g} = \left(\sum_{m: k_m > 0} |\mathbf{e}_m\rangle \langle \mathbf{e}_m| \right) \otimes |g\rangle \langle g|, \quad \Pi_{\leftarrow, 1g} = \left(\sum_{m: k_m < 0} |\mathbf{e}_m\rangle \langle \mathbf{e}_m| \right) \otimes |g\rangle \langle g|. \quad (47)$$

Then the source's observables are exactly

$$T_{1g}(t) = \langle \psi(t) | \Pi_{\rightarrow, 1g} | \psi(t) \rangle = \sum_{m: k_m > 0} |C_{\mathbf{e}_m, g}(t)|^2, \quad R_{1g}(t) = \sum_{m: k_m < 0} |C_{\mathbf{e}_m, g}(t)|^2. \quad (48)$$

These are `p_transmitted` and `p_reflected` in `local_project/simu_Fock_space/src/experiment.py:143--167`. They are *joint* bare-TLS-ground, exactly-one-field-photon sector probabilities at one finite time. The same function reports the total norm, bare P_e , and p_{ν} of Eq. (40). This is substantially nearer to a scattering question than F_a or Q , because it asks about the side and photon count of one calculation. It is still not the complex S -matrix entry of Eq. (43).

Assuming every one-photon mode has either positive or negative k , one has the exact finite-basis identity

$$T_{1g}(t) + R_{1g}(t) = p_{1g}(t), \quad p_{1g} = \sum_m |C_{\mathbf{e}_m, g}|^2, \quad (49)$$

not generally $T_{1g} + R_{1g} = 1$. If the global state is normalized, the omitted probability is

$$1 - T_{1g} - R_{1g} = p_{1e} + \sum_{\nu \neq 1} (p_{\nu g} + p_{\nu e}), \quad (50)$$

where $p_{\nu s} = \sum_{\mathbf{n}: \nu(\mathbf{n})=\nu} |C_{\mathbf{n}s}|^2$. This partition includes temporarily excited TLS, vacuum components, and multiphoton components. Under RWA with exactly one conserved total bare excitation and only nonzero signed modes, it simplifies to $T_{1g} + R_{1g} + P_e = 1$ at every time. No such simplification holds without RWA. At an appropriately late time, vanishing *excess* local excitation and a separated packet are evidence for asymptotic interpretation, but the dressed ground itself can retain bare TLS excitation and local virtual photons.

The local code's position-space routine forms a Fourier series from the one-photon ground-sector amplitudes (`local_project/simu_Fock_space/src/experiment.py:169--180`). This can show whether the outgoing packet has left the TLS and whether periodic wraparound has begun. Its amplitudes include an explicit $e^{-i|k_m|t}$ factor because that implementation's propagated coefficients use its own picture convention. The audited QuTiP calculation stores Schrödinger-picture coefficients directly. Any comparison of *complex* wavefunctions or spectral amplitudes must align those picture conventions before judging phases.

15 A rigorous bridge: what a fidelity can bound about T and R

Suppose two *normalized* outgoing kets $|\psi_A\rangle, |\psi_B\rangle$ have been embedded into the same physical Hilbert space, with the same signed mode grid and the same measurement projector $\Pi_{\rightarrow, 1g}$. Write $\rho_A = |\psi_A\rangle\langle\psi_A|$ and $\rho_B = |\psi_B\rangle\langle\psi_B|$. The trace distance of two pure states is

$$D(\rho_A, \rho_B) = \frac{1}{2} \|\rho_A - \rho_B\|_1 = \sqrt{1 - |\langle\psi_A | \psi_B\rangle|^2} = \sqrt{1 - F_s}. \quad (51)$$

For every effect $0 \leq E \leq I$, in particular any projector, the variational characterization of trace distance gives $|\text{Tr}[E(\rho_A - \rho_B)]| \leq D$. Therefore

$$\boxed{|T_{1g}^A(t) - T_{1g}^B(t)| \leq \sqrt{1 - F_s(t)}, \quad |R_{1g}^A(t) - R_{1g}^B(t)| \leq \sqrt{1 - F_s(t)}}. \quad (52)$$

This is the precise bridge from a *correct* global fidelity to the two directional probabilities. For example, $F_s(t_*) \geq 0.99$ guarantees that each of these probability differences at the *same* t_* is at most 0.1; a much tighter fidelity is needed for percent-level probability accuracy. The inequality compares two finite computations. It cannot show that either computation is close to the physical continuum answer without a validated reference or convergence argument. It also does not bound the error of a complex $t(k)$ from one integrated packet probability.

At matching output times, concavity of the square root gives an average statement:

$$\frac{1}{J} \sum_{j=0}^{J-1} |T_A(t_j) - T_B(t_j)| \leq \frac{1}{J} \sum_j \sqrt{1 - F_s(t_j)} \leq \sqrt{1 - \bar{F}_s}. \quad (53)$$

But Eq. (53) controls only a *mean* probability difference. It supplies no useful endpoint bound when the endpoint is one of many samples. This is why the coupling-sweep mean cannot certify an outgoing transmission result by itself. The full $F_s(t_j)$ trace, or at least F_s in a chosen outgoing-time window, is required.

For the TLS probability, trace distance after partial trace and the fidelity–trace-distance inequality give

$$|P_e^A(t) - P_e^B(t)| \leq D(\rho_a^A(t), \rho_a^B(t)) \leq \sqrt{1 - F_a(t)}. \quad (54)$$

This is the natural meaning of a high **atom** fidelity: the two simulations agree on *TLS measurements*. It says nothing direct about T_{1g} or R_{1g} , because their projectors act on both TLS and field.

The exact field fidelity F_f similarly controls a *field-only* direction projector, for example $(\sum_{m:k_m>0} |e_m\rangle\langle e_m|) \otimes I_a$, with probability $T_{1g} + T_{1e}$. It does not alone bound T_{1g} , which also selects the TLS ground state. The quotient Q has *no* analogue of Eqs. (52)–(54). To see this concretely, reinterpret counterexample (b) of Section 8 with field states $|0\rangle = |1_R\rangle$, $|1\rangle = |1_L\rangle$. It has $Q = 1$ but $T_{1g}^A = 1$ and $T_{1g}^B = 1/4$. Thus a perfect **photon_bound** value can coexist with a 3/4 transmission-probability disagreement.

Nor does the converse of Eq. (52) hold: equal T and R need not imply high global or field fidelity. Equation (15) gives $T_A = T_B = 1$ with $F_s = 0$. Integrated directional probabilities discard spectral and phase information. A useful hierarchy is therefore

$$\begin{aligned} \text{full ket agreement} &\implies \text{agreement of identical directional projectors;} \\ &\text{the converse requires extra assumptions.} \end{aligned} \quad (55)$$

16 How to extract the quantities relevant to a photon probe

16.1 First target: finite-time, sector-resolved probabilities

For any saved state in the audited repository, the *mathematical* operation corresponding to Eq. (48) is to inspect the amplitudes with exactly one waveguide photon and the bare TLS ground label:

$$b_m(t_j) = C_{e_m,g}(t_j), \quad T_{1g}(t_j) = \sum_{k_m>0} |b_m(t_j)|^2, \quad R_{1g}(t_j) = \sum_{k_m<0} |b_m(t_j)|^2. \quad (56)$$

In the `StateTruncated` and `StateTotalCap` wrappers, a label `\{'n3':1,'atom':'g'\}` denotes $|e_3;g\rangle$; omitted `n` keys mean zero (`src/states.py:43--53,66--87,140--167,232--248`). The `StateAtom` label also needs `'n_atom':0`. The sums must use `config.frequencies[m]` *after* grid construction, not the requested mode count. One may compute these from already stored QuTiP ket vectors after wrapping them in the matching `State` class, as `scripts/sweep_g_fidelity.py:54--57` already does for fidelity calculation. No global density matrix is needed.

The output should contain $T_{1g}(t)$, $R_{1g}(t)$, $P_e(t)$, $p_\nu(t)$, and the initial left-moving weight of Eq. (11). Plotting these against the *actual sim.times* distinguishes an incoming stage, interaction stage, outgoing plateau and periodic recurrence. A plateau in T_{1g}, R_{1g} together with a spatially separated packet is a necessary practical sign of an outgoing measurement window; its absence makes a single final number timing-dependent. For **rabi-truncation**'s default packet, first enforce a right-moving input or explicitly account for the initial $k < 0$ amplitude. Merely subtracting $R_{1g}(0)$ from $R_{1g}(T)$ is not generally a valid scattering probability, because initially left-going and newly reflected amplitudes can interfere in the same final modes. Coherent channel preparation is the clean solution.

16.2 Second target: elastic spectral amplitudes

If only weak dressing is relevant and the bare one-photon ground sector is an adequate channel approximation, inspect the *complex* numbers $b_m(t)$, not only $|b_m(t)|^2$. For an outgoing time t_* , a naive interaction-picture coefficient is

$$\tilde{b}_m(t_*) = e^{i(\omega_m + E_{\text{ch}})t_*} b_m(t_*), \quad (57)$$

where $E_{\text{ch}} = 0$ for the bare ground convention of Eq. (7), but should be the dressed ground energy for an exact channel construction. The phase factor removes the known free propagation. For each $\kappa > 0$ for which both signs occur on the grid, let $m_+(\kappa)$ and $m_-(\kappa)$ be the unique mode indices satisfying $k_{m_+(\kappa)} = \kappa$ and $k_{m_-(\kappa)} = -\kappa$; write $b_{\kappa,\pm} = b_{m_\pm(\kappa)}$, $\tilde{b}_{\kappa,\pm} = \tilde{b}_{m_\pm(\kappa)}$, and $c_{\kappa,+} = c_{m_+(\kappa)}$. If the input has $c_m = 0$ for $k_m < 0$, and if the packet and channel approximations justify elastic diagonal scattering, one may estimate

$$t(\kappa) \simeq \frac{\tilde{b}_{\kappa,+}(t_*)}{c_{\kappa,+}}, \quad r(\kappa) \simeq \frac{\tilde{b}_{\kappa,-}(t_*)}{c_{\kappa,+}}, \quad \kappa > 0, \quad (58)$$

only at modes with appreciable incoming amplitude $|c_{\kappa,+}|$ and with a matched $\pm\kappa$ pair. Reflection phases depend on the coordinate origin and x_a ; shifting the scatterer by x_a multiplies a reflected plane-wave coefficient by a convention-dependent factor such as $e^{2i\kappa x_a}$. A free-propagation run at $g = 0$ is a practical phase reference. Dividing by a tiny $c_{\kappa,+}$ is ill-conditioned; a narrowband packet cannot recover $t(\kappa)$ far outside its spectral support.

Equation (58) is *not* an exact recipe in the ultrastrong no-RWA regime because $|\mathbf{e}_m; g\rangle$ is not an exact dressed asymptotic channel. For quantitative S -matrix work there, define $|G\rangle$ and the in/out channel injection in Eq. (42), or measure outgoing localized wavepacket amplitudes sufficiently far from the interacting region and compare against free propagation. Resolve inelastic and bound-state channels separately. The finite bare projections remain useful diagnostics along the way, but their limitations must travel with any reported t, r .

16.3 Minimal comparison that bridges the two repositories

To compare their observables at a fixed finite resolution, first make the two programs represent the *same* finite model:

1. Use the same signed $\{k_m\}$ and the same $\omega_m = c|k_m|$, including mode order and zero-mode policy. The user code's IR/UV band and the audited FFT-like grid otherwise select different Hamiltonians. Both positive and negative resonant modes must be present to measure reflection.
2. Match L , x_a , ω_a , photon packet k_0, x_p , and width. Comparing Eq. (9) with the local Gaussian $e^{-(k-k_0)^2/(4\sigma_k^2)}$ gives $\sigma_k = 1/(\sqrt{2}\sigma_p)$ for equal amplitude envelopes. Set the negative- k coefficients to zero in both preparations and normalize on the retained positive modes.
3. Match the coupling *matrix elements*, including any $L^{-1/2}$ scaling and phase convention. Matching the symbols g and D by name is insufficient; for the same $\sqrt{|k|}$ profile a candidate relation is $g = D/\sqrt{L}$ in the conventions of the two implementations, followed by an explicit matrix-element check.
4. Use the same Fock projector P_N and RWA switch. The local integrator's work basis and post-step projection can differ from exact unitary evolution with $P_N H P_N$; any residual difference must be separated from solver step-size effects. Compare initial vectors after embedding: $F_s(0)$ should be one for the one-photon preparation.
5. At the same physical output times, compare norm convention, $P_e(t)$, $p_\nu(t)$, $T_{1g}(t)$, $R_{1g}(t)$ and the spatial packet. Then use $F_s(t)$ and Eq. (52) as a cross-check of those probabilities. A field fidelity from Eq. (27) can diagnose field disagreement that F_a hides.
6. Identify an outgoing window before periodic wraparound and repeat it for increasing N , momentum bandwidth, and L . Only after those checks use Eq. (58) for tentative complex amplitudes, or the dressed-channel definition for a controlled non-RWA S matrix.

This sequence separates three logically different claims: *code agreement* at fixed finite model, *convergence* of that model as regulators are varied, and *scattering interpretation* of the late-time channel projections. A fidelity curve addresses the first, and only indirectly the second and third.

17 Interpretation guide for a plotted curve

Given any figure labelled “fidelity versus g ”, the following equations are the shortest reliable route to its meaning:

$$\begin{aligned}
 \text{state:} & \quad \overline{\left| \sum_{\mathbf{n},s} (C_{\mathbf{n}s}^X)^* C_{\mathbf{n}s}^{\text{ref}} \right|^2}, \\
 \text{atom:} & \quad \overline{\left[\text{Tr} \sqrt{\sqrt{\rho_a^X} \rho_a^{\text{ref}} \sqrt{\rho_a^X}} \right]^2}, \\
 \text{photon_bound:} & \quad \overline{F_s/F_a} \quad (\text{undefined samples omitted}), \\
 \text{desired bare } T/R: & \quad \sum_{\pm k_m > 0} |C_{e_m,g}^X(t_{\text{out}})|^2, \\
 \text{desired elastic } t/r: & \quad \langle \text{outgoing channel} | S | \text{incoming channel} \rangle.
 \end{aligned}$$

The overbar on the first three lines is the sampled-time mean over the *entire* evolution, while t_{out} is a selected postinteraction time or window. The last two lines are not computed by the principal fidelity sweep. The difference between lines is a difference of *measured operator or channel*, not merely a difference of plotting style.

A concise reading of the saved curves is: agreement of the truncated basis with the finite total-cap reference decreases as the stored coupling g increases; TLS reductions remain much more alike than complete kets; the auxiliary curve adds no independent dynamical channel; coherent initial projections already differ before interaction. This is useful evidence about the chosen Hilbert-space restrictions. It does not establish a transmission spectrum, a reflection spectrum, or an asymptotic non-RWA S matrix. The path from those curves to the target quantity is Eq. (52) for *agreement between two computed T/R values*, followed by the channel and convergence conditions in Sections 13–16 for their physical scattering interpretation.

18 Source map and external physics references

The internal references below identify the exact code paths on which the formulas depend. They are source pointers, not claims that every current test covers every edge case.

Source	Meaning in this note
<code>src/utilities.py:25--184</code>	Configuration defaults, physical mode grid, selection, g , packet center and output-time parameters.
<code>src/states.py:32--167</code> <code>src/states.py:181--358</code>	Occupation sets, dimensions, and label-to-index mappings. Incoming Gaussian, number/coherent preparations, projected normalization, reduced TLS and field matrices.
<code>src/hamiltonians.py:53--88,</code> <code>98--155, 157--272</code> <code>src/simulation.py:27--123</code>	Matrix elements, RWA switch, auxiliary transition factor and selected Hamiltonian. QuTiP ket propagation, time array, P_e , TLS entropy, energy accessors.
<code>src/fidelities.py:8--151</code>	Cross-basis overlap, physical-mode embedding and comparison to one finite reference.
<code>scripts/sweep_g_fidelity.py:</code> <code>43--193</code> <code>scripts/full_cumulative_fidelity.py:19--105</code> <code>scripts/compare_mode_selection.py:75--169</code> <code>src/energy_profile.py:47--146</code>	Reference/candidate choices, three plotted quantities, sampled-time averages and NPZ keys. Neighboring total-cap final-state comparisons. Grid-selection time averages and heatmap quantities. Exact energy partitions, distinct from photon-number probabilities.

Source	Meaning in this note
local_project/simu_Fock_space/ src/experiment.py:119--180	Right-moving input and finite-time T_{1g}, R_{1g} in the independent local implementation.
tests/test_fidelities.py: 21--91, 188--206	Existing fidelity tests; they do not exercise the $ 1, 1; g\rangle$ StateFull versus StateTotalCap failure demonstrated here.

For the continuum and non-RWA interpretation, the following are primary research sources rather than assumptions about the repository: [Sánchez-Burillo et al., *Emergent Causality and the N-photon Scattering Matrix in Waveguide QED*](#) develops asymptotic scattering states and ground-state structure; [Sánchez-Burillo et al., *Scattering in the ultrastrong regime: nonlinear optics with one photon*](#) analyzes dressed bound states and inelastic single-photon scattering; [Gheeraert et al., *Particle Production in Ultra-Strong Coupling Waveguide QED*](#) studies beyond-RWA multiphoton production. Their Hamiltonian implementations differ from this repository; they support the need to specify dressed asymptotic channels and possible inelastic outputs, not a numerical claim about any saved array here.

A Coefficient-matrix derivation of all three fidelities

This appendix derives the comparisons directly from the coefficients in Eq. (1). Fix one common field basis of size B after any required embedding. Define the $B \times 2$ complex matrix

$$C_X = \begin{pmatrix} C_{\mathbf{n}_1 g}^X & C_{\mathbf{n}_1 e}^X \\ \vdots & \vdots \\ C_{\mathbf{n}_B g}^X & C_{\mathbf{n}_B e}^X \end{pmatrix}, \quad \|C_X\|_F^2 = \sum_{i,s} |C_{\mathbf{n}_{is}}^X|^2 = 1. \quad (59)$$

Here $\|\cdot\|_F$ is the Frobenius norm; B is the cardinality of the union of both occupied field bases, with zero rows added for absent occupations. The two columns $c_g^X, c_e^X \in \mathbb{C}^B$ are the field vectors $|\phi_g^X\rangle, |\phi_e^X\rangle$ in a fixed ordering. A source ket vector interleaves them as $(C_{1g}, C_{1e}, C_{2g}, C_{2e}, \dots)$, so $\mathbf{v}[0::2]$ is c_g and $\mathbf{v}[1::2]$ is c_e .

A.1 Trace operations in entries

The field trace is immediate:

$$(\rho_f^X)_{ij} = \sum_{s=g,e} C_{is}^X (C_{js}^X)^*, \quad \rho_f^X = C_X C_X^\dagger. \quad (60)$$

The TLS trace is

$$(\rho_a^X)_{ss'} = \sum_{i=1}^B C_{is}^X (C_{is'}^X)^*, \quad \rho_a^X = C_X^\top C_X^*, \quad (61)$$

where $\top$ means transpose without conjugation and $*$ means entrywise conjugation. The placement of the transpose is worth checking: $C_X^\dagger C_X$ has entry $\sum_i (C_{is}^X)^* C_{is'}^X$, the complex conjugate of Eq. (61); using it as ρ_a would reverse the sign of the off-diagonal imaginary part in the chosen TLS basis. The audited code uses Eq. (61) exactly through its two dot products. Both matrices are positive semidefinite and have trace one because $\|C_X\|_F^2 = 1$.

The global inner product is

$$\langle \psi_A | \psi_B \rangle = \sum_{i,s} (C_{is}^A)^* C_{is}^B = \text{Tr}(C_A^\dagger C_B). \quad (62)$$

Let $G = C_A^\dagger C_B$. Equation (62) says $F_s = |\text{Tr } G|^2$, while Eq. (27) says $F_f = \|G\|_1^2$. The first retains the *signed complex sum* of the two diagonal TLS-overlap channels; the second sums the singular values and optimizes over a TLS unitary in Uhlmann's theorem. The difference between a trace and a trace norm is precisely why reduced-field fidelity can remain one when global states are orthogonal.

A.2 A formula requiring only four complex overlaps

Write

$$G = \begin{pmatrix} \langle \phi_g^A | \phi_g^B \rangle & \langle \phi_g^A | \phi_e^B \rangle \\ \langle \phi_e^A | \phi_g^B \rangle & \langle \phi_e^A | \phi_e^B \rangle \end{pmatrix}. \quad (63)$$

If its singular values are $s_1, s_2 \geq 0$, then $s_1^2 + s_2^2 = \text{Tr}(G^\dagger G)$ and $s_1 s_2 = |\det G|$. Therefore

$$F_f = \text{Tr}(G^\dagger G) + 2|\det G|, \quad F_s = |G_{gg} + G_{ee}|^2. \quad (64)$$

This is an exact alternative to the singular-value call for two TLS states. It uses four complex overlaps and no $B \times B$ field density matrix. It also gives $F_s \leq F_f$ directly from $|\text{Tr } G| \leq \|G\|_1$. The field fidelity still requires a correct physical mode embedding: a formally computed G on mislabeled modes has no physical meaning.

For the TLS, Eq. (21) requires only two 2×2 matrices. Thus none of the three mathematically valid fidelities demands propagation of a global $d \times d$ density matrix. The resource question is how to generate and store the kets and their amplitudes, not whether a mixed global state must be evolved.

B A three-channel worked example: amplitudes, T/R , and fidelities

Consider the RWA one-excitation subspace spanned by three mutually orthogonal bare states:

$$|R\rangle = |1_{k_+}; g\rangle, \quad k_+ > 0; \quad |L\rangle = |1_{k_-}; g\rangle, \quad k_- < 0; \quad |E\rangle = |0; e\rangle. \quad (65)$$

The field vacuum in $|E\rangle$ is orthogonal to both one-photon field states. Let two normalized numerical outputs at a common time be

$$|\psi_X\rangle = u_X |R\rangle + v_X |L\rangle + w_X |E\rangle, \quad |u_X|^2 + |v_X|^2 + |w_X|^2 = 1, \quad X \in \{A, B\}. \quad (66)$$

Every coefficient is a complex number. The finite-time directional and TLS probabilities are

$$T_X = |u_X|^2, \quad R_X = |v_X|^2, \quad P_{e,X} = |w_X|^2, \quad T_X + R_X + P_{e,X} = 1. \quad (67)$$

The last equality follows from the *specific* three-channel subspace, not from generic no-RWA dynamics. The global fidelity is

$$F_s = |u_A^* u_B + v_A^* v_B + w_A^* w_B|^2. \quad (68)$$

The TLS matrix has no off-diagonal term, because the ground-TLS field vector lies in the one-photon sector while the excited-TLS field vector lies in vacuum:

$$\rho_a^X = \begin{pmatrix} T_X + R_X & 0 \\ 0 & P_{e,X} \end{pmatrix}. \quad (69)$$

Substituting in Eq. (21) yields

$$F_a = \left[\sqrt{(T_A + R_A)(T_B + R_B)} + \sqrt{P_{e,A} P_{e,B}} \right]^2. \quad (70)$$

The TLS fidelity depends on the *sum* $T + R$ but not on how it splits between directions. If both TLS populations agree, $P_{e,A} = P_{e,B} = p$, then $F_a = [(1 - p) + p]^2 = 1$ for *any* two directional splits. In particular, at an ideal outgoing time $p = 0$, $F_a = 1$ even if $T_A = 1, R_A = 0$ and $T_B = 0, R_B = 1$. This is the exact mechanism behind the warning about the **atom** sweep curve.

The same probabilities can also hide phases. Take $w_A = w_B = 0$, $u_A = v_A = 1/\sqrt{2}$, $u_B = 1/\sqrt{2}$, $v_B = -1/\sqrt{2}$. Both outputs have $T = R = 1/2$, but Eq. (68) gives $F_s = 0$. Their relative transmitted/reflected phase differs by π . A probe of *only* T, R cannot detect this; an interference experiment or complex channel-amplitude reconstruction can. A global fidelity can detect it but still cannot tell which phase difference is physically correct without a reference.

The matrix G of Eq. (63) is in this example

$$G = \begin{pmatrix} u_A^* u_B + v_A^* v_B & 0 \\ 0 & w_A^* w_B \end{pmatrix}, \quad F_f = (|u_A^* u_B + v_A^* v_B| + |w_A^* w_B|)^2. \quad (71)$$

The absolute values in F_f permit a different relative TLS phase from the global overlap. This worked calculation exhibits all three metric constructions using only three physical channels.

C Why incident direction and phase reference matter for an S column

At a fixed positive energy ω with two degenerate propagation directions, assemble incoming and *elastic* outgoing channel amplitudes into two-component vectors $c = (c_+, c_-)^T$ and $b = (b_+, b_-)^T$. The elastic part of the S matrix acts as

$$\begin{pmatrix} b_+ \\ b_- \end{pmatrix} = \begin{pmatrix} S_{++} & S_{+-} \\ S_{-+} & S_{--} \end{pmatrix} \begin{pmatrix} c_+ \\ c_- \end{pmatrix}, \quad (72)$$

Other outgoing channels, including inelastic multiphoton channels, are orthogonal labels in the full S matrix and must be accounted for separately. In an ideal elastic reciprocal and left-right symmetric model one often has $S_{++} = S_{--} = t$ and $S_{+-} = S_{-+} = r$, but the audited finite code does not impose that reduced form as a measurement assumption.

If $c_- = 0$, Eq. (72) isolates one elastic matrix column: $b_+ = S_{++}c_+$ and $b_- = S_{-+}c_+$. If $c_- \neq 0$, then $b_- = S_{-+}c_+ + S_{--}c_-$. Squaring gives the interference term

$$|b_-|^2 = |S_{-+}c_+|^2 + |S_{--}c_-|^2 + 2\Re[(S_{-+}c_+)^*(S_{--}c_-)]. \quad (73)$$

This makes the logical problem with an initially bidirectional Gaussian exact: the final left-moving probability is neither $|r|^2$ nor $|r|^2 + P_{\leftarrow}^{\text{in}}$ in general. Even subtracting the initial left-moving weight does not remove the interference term. A purely right-moving input, or multiple independently prepared inputs sufficient to invert Eq. (72), is needed to identify one S -matrix column.

For one weakly dressed finite calculation, a free reference run defines a more transparent phase convention than Eq. (57). Let $b_m^{(g)}(t)$ be a complex bare-channel coefficient with the TLS coupling on, and let $b_m^{(0)}(t) = e^{-i\omega_m t} c_m$ be the uncoupled output from the *same* initial packet and same code picture. Where $c_m \neq 0$, the ratio $b_{m,+}^{(g)}(t_*)/b_{m,+}^{(0)}(t_*)$ cancels the free propagation phase and estimates the elastic transmission coefficient under the assumptions stated in Section 16. For reflection, the uncoupled negative-direction coefficient vanishes if the input was properly one-sided; divide the outgoing negative coefficient by the *positive* free input coefficient at the matched $|k|$, then account for the spatial origin. This is a phase-sensitive operation. It cannot be replaced by a fidelity with the free-evolved packet because one scalar overlap integrates over all excited modes and both directions.

With unequal incoming and outgoing group velocities, probabilities associated with unit *flux* carry velocity or density-of-states factors. The simple $|t|^2, |r|^2$ formulas of Eqs. (44)–(46) rely on the stated linear, even, equal-speed channels. Any numerical scattering coefficient should declare its channel normalization as well as its phase origin.

D Source-to-equation audit of the coupling-sweep pipeline

The following listing excerpts are read directly from the audited Python files at LaTeX compilation. They keep the exact line numbers visible, so the mathematics above can be traced without reconstructing the control flow from a plot filename.

D.1 Which basis labels are summed?

```

8 def common_basis(state1: states.State, state2: states.State):
9     """Return the basis class to iterate over when computing the inner product.
10
11     Requirement: every label yielded by `<class>.all_states()` must index correctly
12     into both state1 and state2. StateTruncated labels ({'n_m': k, 'atom': s})
13     embed unambiguously into Full / Atom (with n_atom=0) / TotalCap (other modes=0),
14     so it is the universal common subspace for heterogeneous comparisons.
15     """
16     known = {states.StateFull, states.StateTruncated, states.StateAtom, states.StateTotalCap}
17     t1, t2 = type(state1), type(state2)

```

```

18     if t1 not in known:
19         raise Exception(f"Unknown state type: {t1}")
20     if t2 not in known:
21         raise Exception(f"Unknown state type: {t2}")
22     if t1 is t2:
23         return t1
24     return states.StateTruncated

```

The returned class is the same class when types agree; otherwise it is `StateTruncated`. This determines the index set in Eq. (13). The word `isinstance` in lines 35–40 below is true for the concrete class and its mixins; when neither object is a truncated state, the code constructs a minimal truncated object for label enumeration. No new physical state is propagated by this object.

```

26 def fidelity_statevector(state1: states.State , state2: states.State) -> float:
27     """
28     :param state1 in a basis:
29     :param state2 in another basis (can be same):
30     :return: The fidelity between the two states
31     """
32
33     common = common_basis(state1, state2)
34
35     if isinstance(state1, common) and isinstance(state2, common):
36         ref = state1 if state1.compute_dim() <= state2.compute_dim() else state2
37     elif isinstance(state1, common):
38         ref = state1
39     elif isinstance(state2, common):
40         ref = state2
41     else:
42         # Neither state is in the common basis class: construct a minimal
43         # instance just for basis iteration (all_states / state_to_index only
44         # depend on .config).
45         ref = common.__new__(common)
46         ref.config = state1.config
47
48     fidelity = 0
49     for basis_element in ref.all_states():
50         fidelity += np.conj(state1[basis_element]) * state2[basis_element]
51
52     fidelity = np.abs(fidelity) ** 2
53     return fidelity

```

Line 49 enumerates labels; line 50 reads their two amplitudes and adds their complex product; line 52 squares the modulus *after* summation. This procedure is exact when the enumerated set equals the nonzero support intersection and each label denotes the same physical k modes. The $M = 2, N = 2$ `StateFull-StateTotalCap` counterexample in Section 5 violates the first condition.

D.2 Why the second fidelity is 2×2

```

228 def atom_density_matrix(self) -> np.ndarray:
229     """2x2 reduced density matrix of the atom, tracing over photon degrees of freedom.
230
231     All bases store atom as the lowest index bit (idx*2 + atom_index), so
232     v[0::2] = ground amplitudes, v[1::2] = excited amplitudes for any basis.
233     Rows/cols ordered as [g, e].
234     """
235     v_g = self.v[0::2]
236     v_e = self.v[1::2]
237     return np.array([
238         [v_g @ v_g.conj(), v_g @ v_e.conj()],
239         [v_e @ v_g.conj(), v_e @ v_e.conj()]
240     ])

```

For $v = (C_{1g}, C_{1e}, \dots)$, lines 235–236 form c_g, c_e ; lines 237–240 form exactly Eq. (61). No information about the sign of a photon’s wavevector survives this partial trace.

```

66 rho1 = qutip.Qobj(s1.atom_density_matrix())
67 rho2 = qutip.Qobj(s2.atom_density_matrix())
68 return float(qutip.fidelity(rho1, rho2)) ** 2
69
70
71 def fidelities_over_time(states_a, states_b):
72     """Return (F_state, F_atom, F_photon_bound) time series."""
73     f_state = np.array([fidelity_statevector(a, b) for a, b in zip(states_a, states_b)])
74     f_atom = np.array([fidelity_atom(a, b) for a, b in zip(states_a, states_b)])
75     # Avoid div-by-zero: when atom fidelity collapses, the bound is undefined.
76     with np.errstate(divide="ignore", invalid="ignore"):
77         f_photon = np.where(f_atom > 0, f_state / f_atom, np.nan)
78     return f_state, f_atom, f_photon

```

Lines 66–67 build ρ_a^A, ρ_a^B ; line 68 squares QuTiP’s root fidelity. Lines 73–74 compute state and atom fidelities at each paired saved time. Line 77 evaluates the quotient only when its denominator is positive. The return statement at line 78 names that quotient `f_photon`, but its algebra is Eq. (23), not Eq. (27).

```

81 def sweep(base_config: Config, g_values: np.ndarray):
82     """Returns dict[scheme] -> dict[metric] -> array over g."""
83     metrics = ["state", "atom", "photon_bound"]
84     out = {scheme: {m: np.empty_like(g_values, dtype=float) for m in metrics}
85            for scheme in SCHEMES}
86
87     for i, g in enumerate(g_values):
88         print(f"\n[{i+1}/{len(g_values)}] g = {g:.4g}")
89         ref_states = wrap_states(run(replace(base_config, g=float(g), truncation=REFERENCE)))
90
91         for scheme in SCHEMES:
92             sch_states = wrap_states(run(replace(base_config, g=float(g), truncation=scheme)))
93             f_state, f_atom, f_photon = fidelities_over_time(sch_states, ref_states)
94             out[scheme]["state"][i] = float(np.mean(f_state))
95             out[scheme]["atom"][i] = float(np.mean(f_atom))
96             out[scheme]["photon_bound"][i] = float(np.nanmean(f_photon))
97             print(f"    {scheme:<16} state={out[scheme]['state'][i]:.6f}  "
98                   f"atom={out[scheme]['atom'][i]:.6f}  "
99                   f"photon~={out[scheme]['photon_bound'][i]:.6f}")

```

Lines 89 and 92 construct fresh reference/candidate runs at the same g . Lines 94–96 take three distinct means; the last is a mean of pointwise quotients. No source line in this pipeline constructs an outgoing projector of Eq. (47), a field Gram matrix of Eq. (26), or an asymptotic channel amplitude of Eq. (43).